

LORETO GIRLS 2025 PREDICTION



(EXPECTED KCSE EXAM)



SCHOOL..... INDEX NO.....

SIGN.....

Kenya Certificate of Secondary Education (K.C.S.E)

233/3

CHEMISTRY

PAPER 3

PRACTICAL

TIME: 2 ¼ HOURS

INSTRUCTIONS TO CANDIDATES

- [a] Answer ALL questions in the spaces provided in each question.
- [b] Mathematical tables and electronic calculators may be used for calculations.
- [c] all working must be clearly shown where necessary.

FOR EXAMINERS ONLY

QUESTION	MAXIMUM SCORE	CANDIDATES SCORE
1	15	
2	13	
3	12	
Total	40	

1. You are provided with
 - 2.0M NaOH solution labelled B
 - Sulphuric(VI) acid solution labelled A

You are to:

- [a] Prepare a dilute solution of NaOH solution.
- [b] Determine the concentration of in moles per litre.

PROCEDURE 1

- i. Using a pipette 25.0cm³ of solution B and place it into 250cm³ volumetric flask.
- ii. Add about 200cm³ of distilled water and shake well.
- iii. Add more water to make up to 250cm³ mark. Label this solution C

- [a] Calculate the concentration of the dilute solution C in moles per litres. **[2mks]**

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PROCEDURE 2

- i. Fill the burette with solution A and record the readings in the table below.
- ii. Pipette 25cm³ of dilute solution C and place it into 250ml conical flask.
- iii. Add 2-3 drops of phenolphthalein indicator.
- iv. Titrate with solution A.
- v. Record your results in the table below.
- vi. Repeat the titration two or more times and complete the table.

	I	II	III
Final burette reading (cm ³)			
Initial burette reading (cm ³)			
Volume of solution A (cm ³)			

[4mks]

- [a] Determine average volume of the acid (solution A) used. **[1mk]**

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[b] Determine moles of dilute solution C in the volume used. [2mks]

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[c] Write an equation for the reaction taking place. [1mk]

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[d] Determine the number of moles of A used. [2mks]

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[e] Determine the concentration of A in moles per litre. [2mks]

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2. You are provided with the following

- 2M sodium hydroxide solution, solution B
- 2M hydrochloric acid, solution D

You are required to determine the molar enthalpy of neutralization of the acid using sodium hydroxide.

PROCEDURE

[i] Measure out 20cm³ of acid into a clean plastic beaker.

[ii] Record the temperature of this solution in the table below

[iii] Measure 5cm³ of sodium hydroxide and add it to the hydrochloric acid.

[iv] Stir with the thermometer and record the maximum temperature reached.

[v] Repeat the above procedure adding 5cm³ portions of sodium hydroxide until the total volume of the solution is 50cm³.

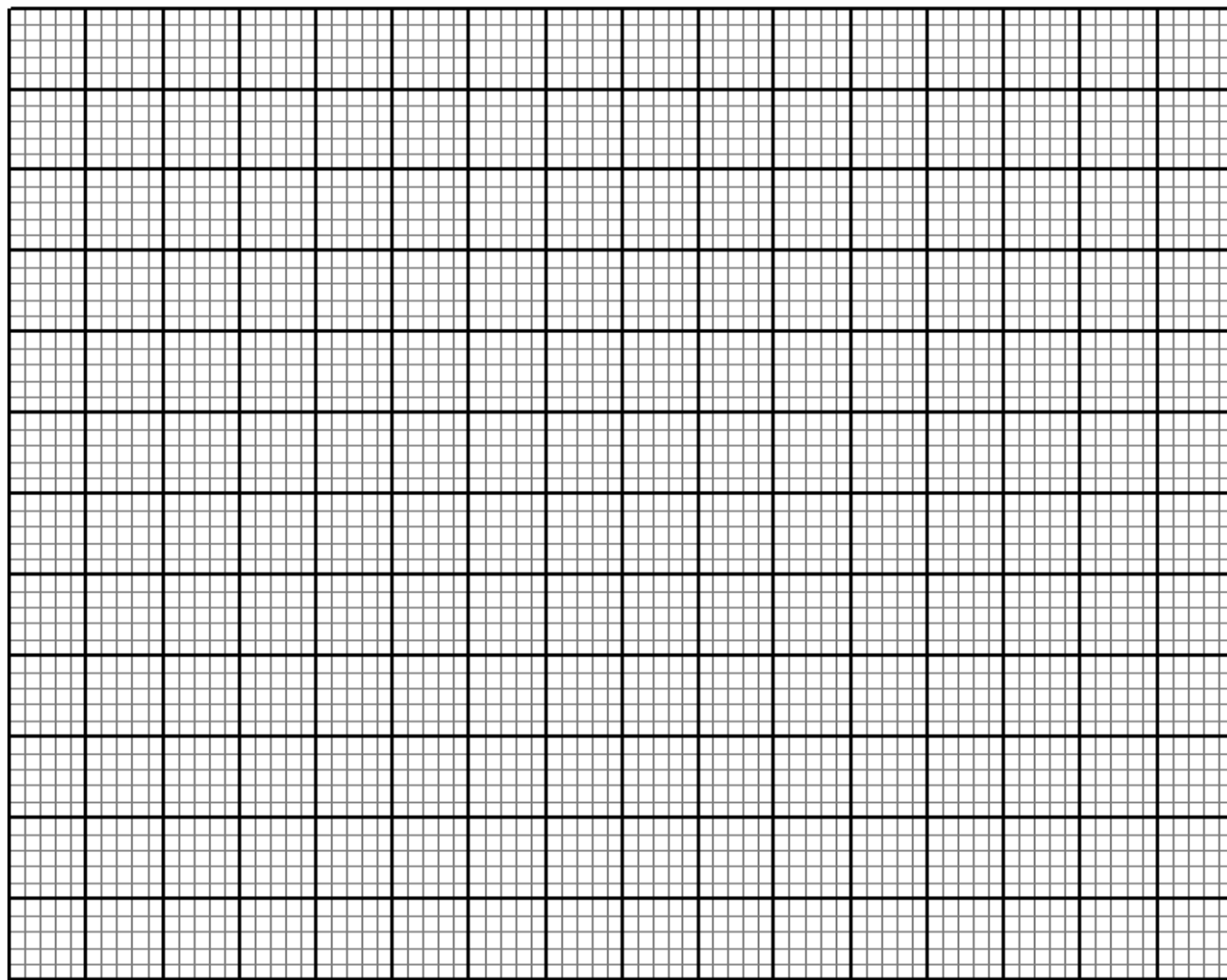
Volume of acid(cm ³)	20	20	20	20	20	20	20
Volume of NaoH added cm ³	0	5	10	15	20	25	30
Temperature(°C)of solution							

[3mks]

You are required to:

[a]. Plot a graph of temperature rise against sodium hydroxide added.

[3mks]



[b] From your graph determine:

[i] maximum temperature change.

[1mk]

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[ii] the volume of NaOH that is required for complete neutralization

[1mk]

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[iii] Calculate the molar enthalpy of neutralization for this reaction. ($C=4.2\text{J/g/K}$) assume density of solution is 1gcm^{-3})

[2mks]

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[iii] The theoretical molar heat of neutralization is -57.2kJ/mol . Compare your value in [ii] above with the theoretical value. Give the reasons for any differences noted between these two values.

[2mks]

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3. You are provided with solid N carry out the tests below and record your observations and inferences.

[a] Place a spatula of N in a test tube and add 5cm^3 of water and shake well divide the solution in to three portions.

OBSERVATION (1mk)	INFERENCE (1mk)

[b] Add sodium hydroxide to the first portion drop wise while observing till in excess

OBSERVATION (1mk)	INFERENCE (2mks)

[c] Add ammonia solution to the second portion drop wise until in excess.

OBSERVATION (1mk)	INFERENCE (1mk)

[d] Add four drops of potassium iodide solution to the third portion.

OBSERVATION (1mk)	INFERENCE (1mk)

[e] Add three drops of acid barium nitrate to the fourth followed by 5 drops of nitric acid.

OBSERVATION (2mks)	INFERENCE (1mk)